

Physics Midterm · Practice Test 1

Physics · vectors and scalars, 1-D and 2-D kinematics, Newton's laws of motion

Name: _____

Date: _____

Period: _____

Time: 50 minutes. **Total:** 80 points. Part A: 17 multiple-choice questions, 2 points each (34). Part B: 5 free-response questions (46). In Part B show your work: write the equation, substitute with units, then give the answer with units (and direction for a vector).

Reference. $g = 9.81 \text{ m/s}^2$. Constant acceleration: $v = u + at$ · $s = ut + \frac{1}{2}at^2$ · $v^2 = u^2 + 2as$ · $s = \frac{1}{2}(u + v)t$. Components: $A_x = A \cos \theta$, $A_y = A \sin \theta$ (θ measured counter-clockwise from the +x axis). Newton's second law: $F_{\text{net}} = ma$. Weight: $W = mg$. Up and right are positive unless a question says otherwise. A calculator may be used (degree mode).

Part A · Multiple Choice (2 points each; circle one answer)

- Which quantity is a vector?
A speed
C displacement
B mass
D time
- Which list contains only scalars?
A distance, speed, mass
C force, distance, time
B velocity, time, mass
D acceleration, speed, energy
- Two forces act on an object: 6.0 N east and 8.0 N north. What is the magnitude of the resultant?
A 2.0 N
C 48 N
B 14 N
D 10 N
- A velocity of 20 m/s is directed 30° above the horizontal. What is its horizontal component?
A 10 m/s
C 20 m/s
B 17.3 m/s
D 34.6 m/s
- A runner goes once around a 400 m track and stops where she started. Her distance and displacement are
A 400 m and 400 m
C 400 m and 0 m
B 0 m and 400 m
D 0 m and 0 m
- A car's velocity changes from 10 m/s to 30 m/s in 5.0 s. What is its acceleration?
A 2.0 m/s^2
C 6.0 m/s^2
B 4.0 m/s^2
D 8.0 m/s^2
- The slope of a velocity–time graph gives the
A displacement
C distance
B speed
D acceleration
- A ball is thrown straight up. At its highest point, its velocity and acceleration are
A zero and zero
C 9.81 m/s upward and zero
B zero and 9.81 m/s^2 downward
D zero and 9.81 m/s^2 upward

Part B · Free Response (show your work)

18. Vectors and scalars (10 points)

(a) Write "scalar" or "vector" for each quantity. (3 points)

distance	displacement	speed	velocity	mass	weight

(b) A force of 50 N acts at 37° above the +x axis. Find its x- and y-components. (2 points)

Work:

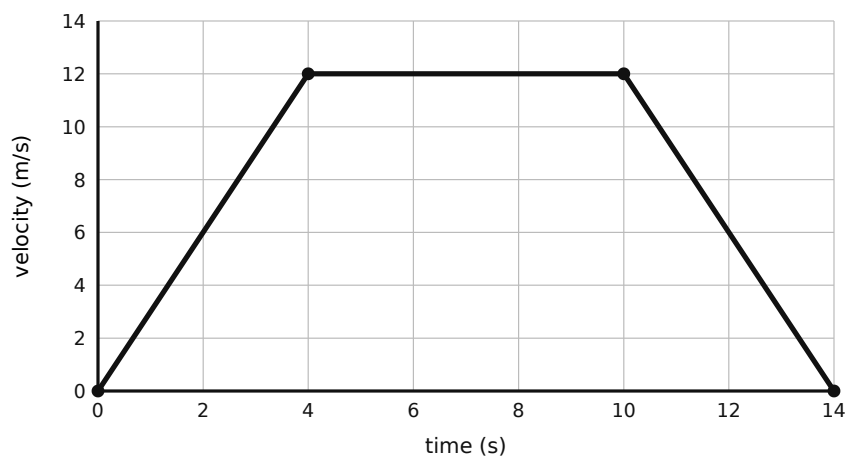
(c) A hiker walks 3.0 km east and then 4.0 km north. (i) Draw a tip-to-tail vector diagram that shows both walks and the resultant displacement. (2 points) (ii) Calculate the magnitude and direction of the resultant displacement. (3 points)

(i) Diagram:

(ii) Work:

19. Motion in a straight line (10 points)

The graph shows the velocity of a cyclist moving along a straight road.



(a) Find the acceleration from 0 to 4 s. (2 points)

Work:

(b) Describe the motion from 4 s to 10 s, and state the net force on the cyclist during this time. (2 points)

(c) Find the acceleration from 10 s to 14 s. (2 points)

Work:

(d) Find the total displacement from 0 to 14 s. (2 points)

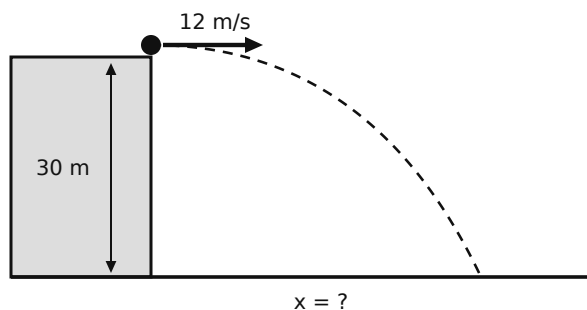
Work:

(e) A ball is thrown straight up at 19.6 m/s. Find the time it takes to reach its highest point and the maximum height. (2 points)

Work:

20. Projectile motion (10 points)

A ball is thrown horizontally at 12 m/s from the top of a cliff 30 m high. Ignore air resistance.



(a) How long does the ball take to reach the ground? (3 points)

Work:

(b) How far from the base of the cliff does it land? (2 points)

Work:

(c) Find the vertical component of its velocity just before it hits the ground. (2 points)

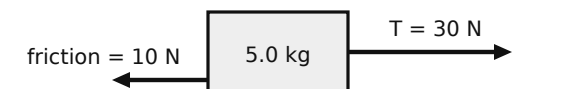
Work:

(d) Find the speed of the ball just before it hits the ground, and the angle its velocity makes below the horizontal. (3 points)

Work:

21. Newton's second law (10 points)

A 5.0 kg crate is pulled along a level floor by a horizontal rope. The tension in the rope is 30 N and a friction force of 10 N acts on the crate.



(a) Draw a free-body diagram of the crate. Label all four forces. (3 points)

Free-body diagram:

(b) Calculate the weight of the crate and the normal force on it. (2 points)

Work:

(c) Find the net force on the crate and its acceleration. (3 points)

Work:

(d) The crate starts from rest. Find its speed and the distance it has moved after 3.0 s. (2 points)

Work:

22. Newton's first and third laws (6 points)

(a) A passenger standing in a bus falls backward when the bus suddenly starts moving forward. Explain why, using Newton's first law. (2 points)

(b) A swimmer pushes backward on the water with her hands. Identify the third-law partner of this force and explain why she moves forward. (2 points)

(c) A student writes: "The weight of a book and the normal force from the table are an action-reaction pair, because they are equal and opposite." Explain why this is wrong. (2 points)
